MILITARY SPECIFICATION

DRUM, AMMUNITION, ROTARY, 7.62MM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of rotary ammunition drum assembly for use on the M28A1 Helicopter Armament Subsystem. The rotary drum assembly can be used for either right-hand or left-hand mounting and shall consist of an inner and outer drum, gear reduction drive, rounds counter switch, last round switch, and exit chute for supplying 7.62mm ammunition to the M134 Machine Gun (GAU-2B/A).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military		
MIL-W-13855	-	Weapons, Small Arms and Aircraft Armament Subsystem, General Specification for.
MIL-P-14232	-	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of.
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of.
MIL-A-45967	-	Armament Subsystem Helicopter: M28A1

STANDARDS

Military		
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	-	Quality Assurance Terms and Definitions.

FSC 1090

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DRAWINGS

Rock Island Arsenal
J8449901 - Drum, Ammunition Rotary 7.62mm.
IEL8449901 - Index of Inspection Equipment List.

(Product drawings referenced in the specification form a part of Drawing J8449901; Acceptance Inspection drawings referenced in this specification form a part of IEL8449901.)

PUBLICATIONS

Rock Island Arsenal
P8449901 - Drum, Ammunition Rotary 7.62mm, Packaging Data Sheet.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.2). Unless otherwise specified (see 6.2) the first article shall include the pilot pack (see 5.1).

3.2 Material, construction and design. The rotary drum shall conform to the material, construction and design requirements specified herein on Drawing J8449901, and in MIL-W-13855.

3.2.1 Storage capacity. The storage capacity of the loaded drum shall be 3,975 ± 25 rounds of 7.62mm ammunition belted with M13 links.

3.2.2 Weight. The weight of the rotary drum without ammunition shall not exceed 65 pounds.

3.3 Functioning requirements. The rotary drum shall be capable of supplying Government standard 7.62mm ball ammunition during firing of the M134 machine gun at the steady state rate of fire of 2,000 ± 300 and 4,000 ± 600 shots per minute (spm) without malfunction or failure. Testing shall be as specified in 4.5.3.2.

3.3.1 Acceleration. The rotary drum shall accelerate from drive motor activation to a cyclic rate of 2,000 ± 300 spm in no more than 0.25 seconds and cyclic rate of 4,000 ± 600 spm in no more than 0.30 seconds.

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3.3.2 Mechanical breakaway torque. The mechanical breakaway torque, at the loading fitting (8449936), required to rotate the drum empty, shall not exceed 9 inch-pounds. The mechanical breakaway torque to load ammunition into the rotary drum shall not exceed 50 inch-pounds, and to unload the ammunition shall not exceed 40 inch-pounds.

3.3.3 Running torque. Torque measured at the torque limiter input shall not exceed the following values under normal operating conditions.

	High Rate 4,000 SPM	Low Rate 2,000 SPM
Start up	20 pound-inches	15 pound-inches
Steady state	9 pound-inches	7 pound-inches

3.3.4 Rounds counter switch. The rounds counter switch shall close to provide electrical continuity between pins E and F of Connector J1 when the rounds counter cam roller passes over a lobe of the rounds counter cam, and shall provide a break in electrical continuity between these same pins when the rounds counter cam roller passes over a depression between lobes. The continuity pulse shall be within 24-30 VDC and at least 15 milliseconds in duration when the drum is being operated at the performance rates specified in 3.3. The rounds counter switch shall be adjusted to the requirements specified on the drawing.

3.3.5 Last rounds switch. The last rounds switch shall return to the switch nonactuated position (electrically open between pins G and H of Connector J1) when the last round has passed the exit chute. The actuator arm shall actuate the switch (electrically close between pins G and H of Connector J1) when there is ammunition in the exit chute. The last rounds switch shall be adjusted to the requirements specified on the drawing.

3.3.6 Torque limiter. The torque limiter shall transmit torque from the flexible shaft to the drive mechanism of the Rotary Drum Assembly. With the output shaft locked the torque limiter shall perform its intended (slipping) function between the limits of 25 to 31 pound-inches.

3.4 Performance requirements.

3.4.1 Endurance. The rotary drum shall effectively support the firing of 28,000 rounds of 7.62mm ammunition with no parts replacements and no stoppages. A stoppage is defined as any malfunction of the rotary drum (excluding ammunition and personnel errors) which prevents the rotary drum from operating as specified herein. Testing shall be as specified in 4.5.3.3.

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3.5 Reliability. When specified in the contract (see 6.2), the rotary drum shall have a minimum acceptable Mean-Rounds-To-Stoppage (MRTS) of at least 75,000 rounds. This MRTS shall be demonstrated at a confidence level of at least 30 percent. (Passing the endurance requirements of 28,000 rounds with no stoppages as specified in 3.4.1 demonstrates a MRTS of 75,000 rounds at 30 percent confidence.) A stoppage is defined as any malfunction of the rotary drum, (excluding ammunition and personnel errors) which prevents the drum from operating as specified herein. Testing for reliability shall be in accordance with 4.4.

3.6 Environmental conditions. When specified in the contract (see 6.2), the rotary drum shall be capable of satisfying the specific environmental conditions in MIL-A-45967. When subjected to the environmental conditions, the rotary drum shall not exhibit any detrimental damage or deterioration which would prevent the drum from operating as specified herein. Testing for environmental shall be in accordance with 4.4.

3.7 Interchangeability. Unless otherwise specified on the drawings, replacement parts shall be interchangeable. Testing shall be as specified in 4.5.3.4.

3.8 Marking. The rotary drum shall be clearly marked in accordance with MIL-W-13855. Each rotary drum shall be identified by a serial number assigned by the procuring activity (see 6.2).

3.9 Workmanship. Workmanship shall be in accordance with MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facility suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

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4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. The first article (initial production unit(s) shall be submitted for inspection in accordance with contract requirements (see 6.2). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein, and when specified in the contract, the reliability test (see 3.5 and 4.6.4) environmental test (see 3.6), and such other inspection as necessary to determine that all requirements of the contract have been met.

4.5 Quality conformance inspection.

4.5.1 Inspection lot. Unless otherwise specified herein, the formation, size and presentation of inspection lots shall be in accordance with MIL-STD-105 and MIL-W-13855. Rotary drums shall be assembled from component parts that have met all inspection requirements specified herein. The inspection lot size and sample size for the endurance test shall be in accordance with the contract (see 6.2).

4.5.2 Examination.

4.5.2.1 Component parts and concurrent repair parts. Examination of parts shall be performed in accordance with the criteria specified in the contract (see 6.2). The contractor's examination of parts shall be accomplished prior to their assembly into the end item or submission for acceptance as repair parts.

4.5.2.2 Rotary drums. Visually and manually examine rotary drums in accordance with the following. The examination provisions shall be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristics by subsequent operations. Reinspections of these characteristics on the completed product is not required provided assurance exist that the characteristics has not been changed, degraded or damaged by subsequent manufacturing, assembling or handling and that adequate inspection records are maintained. Rotary drums failing to meet the requirements shall be rejected.

4.5.2.2.1 Weight. Examine the first three rotary drums for compliance with weight requirement specified in 3.2.2.

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4.5.2.2.2 Each rotary drum shall be visually examined to assure:

a. There are no missing, damaged or incorrectly assembled component parts.
b. Visually examine all marking required by the final assembly drawing to assure they are clear and legible.
c. Visually examine the identification plate for compliance with the applicable drawing, for secure assembly, serialization and to assure that all pertinent data has been applied (see 3.8).
d. Workmanship is in accordance with 3.9.

4.5.3 Testing.

4.5.3.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on a complete rotary drum. Failure of the rotary drum to meet the test requirements specified herein shall be cause for rejection of the rotary drum or the inspection lot of rotary drums. Acceptance shall be deferred and the contractor shall accomplish as applicable the following actions:

a. Conduct a failure analysis study performing a dimensional, physical and visual examination of the components which are suspected to be the cause of failure or malfunction.
b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect(s) in future production.
c. Examine rotary drums, partially assembled rotary drums, and components (including components and subassemblies at in-process or final assembly) to insure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.
d. The Government representative shall verify the results of failure analysis and assure that corrective action, as applicable, has been accomplished prior to submission as a reconditioned lot or reconditioned rotary drum for retest.

4.5.3.2 Functional testing. Each rotary drum shall be tested for functional requirements (see 3.3) using the test methods specified in 4.6.1. Rotary drums failing to meet the functional requirements shall be rejected.

4.5.3.3 Endurance testing. Endurance testing (see 3.4.1 and 4.5.1) shall be as specified in the contract (see 6.2) using test method 4.6.2. Units selected for endurance testing shall have satisfactorily completed all other examinations and tests except interchangeability. Failure of the rotary drum to successfully complete the endurance test shall be cause for rejection of the lot.

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4.5.3.4 Interchangeability testing. A sample of three rotary drums randomly selected from each of the first three months production and from each alternate months production thereafter shall be tested for interchangeability (see 3.7) using the test method specified in 4.6.3. Rotary drums selected for interchangeability shall have been found satisfactory in all other examinations and tests. The rotary drums shall be function tested after interchange of parts. During the function test one rotary drum shall be loaded to storage capacity, and the two remaining rotary drums shall be loaded with 2000 rounds of ammunition. Failure of the interchangeability test shall be cause for rejection of that months production. Prior to submission as a reconditioned lot, the suppliers corrections shall be effected on all rotary drums of the lot and on all rotary drums and component parts not yet accepted. Sample size and test methods for reconditioned lots shall be the same as for the original lot. Failure of any interchangeability test shall also be cuase to perform an interchangeability test on the next two months production.

4.5.3.5 Component parts and concurrent repair parts testing. Raw material testing, part testing, and certification shall be performed in accordance with the criteria specified in the contract (see 6.2). This will include chemical analysis and physical tests of materials, tests of protective finish, heat treatment, bonding, and function of parts as applicable. The contractor shall accomplish these tests prior to assembly of parts into the end item.

4.5.4 Packaging examination and testing. Unless otherwise specified (see 6.2), the packaging examination and testing shall be in accordance with MIL-P-14232.

4.5.5 Inspection equipment.

4.5.5.1 Acquisition, maintenance and disposition. Unless otherwise specified, responsibility for acquisition, maintenance and disposition of acceptance and test equipment prescribed on the List of Inspection Equipment IEL8449901, and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.4.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it must be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.6. Test methods.

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4.6.1 Functioning tests. The rotary drum shall be cycled and fired using live rounds of 7.62mm ammunition. Each rotary drum shall be loaded to storage capacity until three consecutive drums have met the functional requirements (without retest). Thereafter, one drum from each months production shall be loaded to storage capacity. The remaining drums shall be loaded with 2000 rounds of ammunition. Failure to meet any functional requirements except the empty mechanical breakaway torque of 3.3.2, 3.3.3 or 3.3.5 shall be cause to load the next three consecutive drums to storage capacity. The drum shall be tested in accordance with the following paragraphs to determine compliance with the applicable functional requirements.

4.6.1.1 Test set-up. The test equipment required to perform the tests shall include:

a. Power supply - capable to providing a minimum of 20 amperes at 22-30 VDC.
b. Power supply - capable of providing 100 amperes at 22-30 VDC.
c. Torque wrenches with suitable adapters (contractor design) - capable of measuring torques (in-lb) specified in 3.3.2 and 3.3.6.
d. Torque measuring system with suitable adapters (contractor designs) - capable of measuring start-up and steady state torques specified in 3.3.3.
e. Proximity (magnetic) pick-ups - capable of measuring weapon rates of fire specified in 3.3.
f. Chart recorder - capable of recording weapon firing rates (see 3.3), running torque (see 3.3.3), and rounds counter circuit test (see 3.3.4).
g. Ohmmeter - capable of monitoring switch continuity as specified in 3.3.5.
h. Test stand - capable of positioning rotary drum, M134 machine gun and 7.62mm flexible chuting as specified in figure 1.
i. Mount the rotary drum on the test stand utilizing the following parts (see 6.2):

Part	Part number
End Fitting, Chute	8441511
Link, Chute	11690221
Shaft Assembly, Flexible	8441510
Screw, Machine	AN500AD8 (8)
Clamp Assembly, Chute, Support	11690222
Bolt, Machine	MS9122-03 (8)
Adapter Assembly	11691074 (2)
Screw, Machine	11701244 (3)

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Part	Part number
Drive Assembly, Gun 7.62MM	11690280
Chute, Ejection	11690408
Retainer, Link, Cartridge	11690344
Pin, Quick Release	63B10948 (2)
Feeder, Delinking: MAU-56A	11701120
Chute, Ejection	11690407
Stabilizer Assembly	11690474
Machine Gun, 7.62MM, M134	65F9877
Extension, Support	11690229
Extension	11690241
Connector, Plug, Electrical	MS3126F12-10P
Connector, Plug, Electrical	MS3126F-14-5S

4.6.1.2 Mechanical breakaway torque (drum empty) test. The loading fitting shall be rotated in a counter-clockwise direction (facing the exit unit end of the drum) with a torque wrench to determine that the torque required to rotate the drum does not exceed the applicable requirement specified in 3.3.2.

4.6.1.3 Torque limiter test. A torque wrench shall be used to rotate the Torque Limiter's input shaft clockwise (facing the exit unit end of the drum) while restricting the Limiter's output shaft from any movement. The torque required to begin movement of the input shaft shall conform to the value specified in 3.3.6.

4.6.1.4 Storage capacity and torque tests. The rotary drums shall be loaded to storage capacity to determine compliance with 3.2.1 or 2000 rounds on ammunition in accordance with 4.7.1. The ammunition used shall be live 7.62mm ammunition belted with M13 links. During the loading of the rotary drums, the following tests shall be performed.

4.6.1.4.1 Mechanical breakaway torque (loading) test. Using a torque wrench attached to the loading fitting, check the loading torque at a minimum of five approximately equally spaced intervals (including initiation and completion) during ammunition loading to determine that the loading torque does not exceed requirement of 3.3.2.

4.6.1.4.2 Mechanical breakaway torque (unloading) test. Using a torque wrench attached to the loading fitting, check the unloading torque at a minimum of initiation, 25, 50, 75 and 100 rounds while unloading 100 rounds of ammunition from the drum to determine that the unloading torque does not exceed requirement of 3.3.2. Reload the 100 rounds of ammunition.

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4.6.1.5 Firing tests. Firing shall be conducted from the test stand and the following cycles shall be repeated until ammunition is expended. A weapon cooling time of 30 seconds minimum shall be allowed after each cycle. NOTE: Weapon cooling time is based on optimum cooling conditions. If other than optimum cooling conditions exist, cooling time shall be increased as appropriate to prevent weapon damage.

a. A 6 second 4,000 spm burst
b. A 6 second 2,000 spm burst
c. A 4 second 2,000 spm burst
d. A 4 second 4,000 spm burst
e. A 3 second 2,000 spm burst; then without stopping, switch to 4,000 spm for 3 seconds
f. A 3 second, 4,000 spm burst; then without stopping, switch to 2,000 spm for 3 seconds.

During the firing test of the rotary drums, the following tests shall be performed:

4.6.1.5.1 Rounds counter circuit test. A chart recorder shall be attached to pins E and F of connector J1 on the rotary drum to determine if the continuity pulse complies with 3.3.4.

4.6.1.5.2 Running torque test. The torque transducer shall be connected to the torque limiter's input shaft and its output connected to a chart recorder. The resulting chart record shall be evaluated to determine compliance with running torques of 3.3.3.

4.6.1.5.3 Acceleration and rates test. The proximity pick-up shall be located in a position to accurately measure weapon firing rate and its output connected to a chart recorder. The resulting chart record shall be evaluated to determine compliance with firing rates of 3.3 and acceleration times of 3.3.1.

4.6.1.5.4 Last rounds switch test. An ohmmeter shall be connected between pins G and H of connector J1. The resulting continuity check shall be used to determine compliance with 3.3.5.

NOTE: Upon completion of firing test, visually examine rotary drum, flexible chuting and machine gun to assure all ammunition has been expended.

4.6.2 Endurance test. The rotary drum shall be endurance tested on a Government approved test stand that is capable of positioning rotary drum, M134 machine gun and 7.62mm flexible chuting as specified in Figure 1. The rotary drum shall be loaded with a 4,000 round complement of live ammunition. The ammunition shall be fired according to the following duty cycle:

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a. Start the drive motor and run at 2,000 spm for 3 seconds, then run at 4,000 spm for 3 seconds.
b. Pause for 30 seconds minimum.*
c. Start the drive motor and run at 4,000 spm for 4 seconds.
d. Pause for 30 seconds minimum.*
e. Start the drive motor and run at 2,000 spm for 4 seconds.
f. Pause for 30 seconds minimum.*
g. Start the drive motor and run at 4,000 spm for 3 seconds. then run at 2,000 spm for 3 seconds.
h. Pause for 30 seconds minimum.*
i. Repeat a thru h until a total of seven complements (28,000 rounds) of ammunition have been fired.

NOTE: Weapon cooling time is based on optimum cooling conditions. If other then optimum cooling conditions exist, cooling time shall be increased as appropriate to prevent weapon damage.

4.6.2.1 Disposition of endurance tested rotary drums shall be as specified in the contract (see 6.2).

4.6.3 Interchangeability test. The rotary drums shall be tested for interchange of parts by disassembling and reassembling parts as specified below. Interchange of parts shall be accomplished by marking the drums #1, #2, and #3, and recording the serial numbers of each drum. These trays shall be placed in front of each drum to be disassembled and the trays marked as follows:

Drum #1	Drum #2	Drum #3
1A, 1B, 1C	2A, 2B, 2C	3A, 3B, 3C

Each rotary drum shall be disassembled and the parts placed into either Group A, Group B or Group C, according to the group listing of parts below. Interchange and reassemble the trays or parts so that the interchanged drums shall consist of the following groups:

Interchanged	Interchanged	Interchanged
1A, 2B, 3C	1B, 2C, 3A	1C, 2A, 3B

Commercial parts such as screws, nuts, washers, and pins shall be placed in the same tray as their mating or associated part. The commercial parts have not been listed in the group parts list below. Any commercial part rendered unserviceable by disassembly shall be replaced without penalty to the Interchangeability Test. Any noncommercial part rendered unserviceable by disassembly shall be cause for failure of the interchangeability test. After interchange of parts, each reassembled rotary drum shall be function tested in accordance with 4.6.1.

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GROUP A

TORQUE LIMITER ASSEMBLY

8441525	Pin, Straight
8441526	Set Screw
8441531	Washer, Thrust
8441532	Bearing, Sleeve
11691374	Adapter, Flexible Shaft

CHUTE ASSEMBLY

8449965	Pin, Straight
8449969	Chute, Ammunition

DRIVE GEAR ASSEMBLY

8449912	Cover, Magazine - Forward
8449920	Housing, Mechanical Drive (Large Casting)
8449921	Cover, Mechanical Drive
8449933	Gear Cluster, Spur
8449936	Adapter, Hand Loading
8449945	Plate, Bearing Retainer
8449947	Plate, Bearing Retainer
11010124-7	Bearing
11014596-22	Bearings (9)
11691410	Key
11691412	Gear Sub-assembly (1)
11691602	Key

MISCELLANEOUS

8441517	Bracket, Chute Mounting
8449909	Support, Front Drum
8449981	Cable Assembly, Electrical Branched

GROUP B

TORQUE LIMITER ASSEMBLY

8441501	Housing Assembly, Limiter
8441502	Shaft, Shouldered

CHUTE ASSEMBLY

8441546	Spring, Helical Compression
8449962	Spring, Flat
8449966	Knob, Gate Assembly

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GROUP B (Con't)

DRIVE GEAR ASSEMBLY

8449929	Shaft, Intermediate
8449935	Gearshaft, Spur
8449943	Plate, Bearing Retainer
8449948	Housing
8449954	Spacer, Sleeve
8449975	Plate, Bearing Retainer
11010124-5	Bearing, Ball Annular
11010124-6	Bearing, Ball
11014596-13	Bearing
11014596-22	Bearings (3)
11691407	Sprocket Sub-assembly (lower)
11691408	Sprocket Sub-assembly (upper)
11691412	Gear Sub-assembly (1)

MATCHED PAIR

8449928	Gear, Bevel
8449930	Gearshaft, Bevel-Spur

MISCELLANEOUS

8441507	Clip, Spring Tension
8441520	Rod-Handle
8449914	Inner Drum (Less Shaft)
8449922	Cover, Magazine, aft
8449955	Plate, Bearing Retainer
8449977	Bracket, Electric Switch

GROUP C

TORQUE LIMITER ASSEMBLY

8441504	Cap, Torque Limiter
11014596-9	Bearing

TORQUE LIMITER ASSEMBLY

11691365	Washer, Special
11697433-2	Ring, Retaining

MATCHED PAIR

8441503	Housing, Bearing Unit
8441505	Plug, Torque Limiter

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GROUP C (Con't)

CHUTE ASSEMBLY

8449964	Gate, Loading Ammunition

CARRYING HANDLE

8441521	Tube - Handle
8449942	Bushing, Sleeve

DRIVE GEAR ASSEMBLY

8449932	Gear Cluster, Spur
8449934	Gear Cluster, Spur
8449940	Plate, Bearing Retainer
8449941	Plate, Bearing Retainer
8449943	Plate, Bearing Retainer
8449944	Housing, Bearing
8449976	Housing, Bearing
11691369	Shaft, Sprocket (Upper)
11691371	Shaft, Sprocket (Lower)

MATCHED GEAR SET

8449931	Gearshaft, Spur
8449950	Gear, Spur
11010124-8	Bearing

MISCELLANEOUS

8441508	Clip, Spring Tension
8441542	Bearing
8449906	Support, Magazine Drum
8449916	Shaft, Shouldered
8449980	Bracket, Electric Connector
8449999	Drum Assembly, Outer
11689081	Switch, Push

NOTE: The following shims are laminated and may have to be adjusted during the interchange of parts to obtain proper operation and tolerances of the mating parts:

8441516	Shim
8449946	Shim
8449949	Shim

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4.6.4 Reliability test. The reliability test shall be performed by repeating the duty-cycle defined in 4.6.2 to determine compliance with 3.5 for each test cycle. The failure versus test cycles data shall be cumulatively plotted. Testing shall terminate whenever a decision point, as defined in Figure 2, has been reached.

4.6.4.1 Reliability test cycle. One test cycle (representing the functions necessary to perform one mission) shall be accomplished by repeating the duty-cycle of 4.6.2 until a full complement (4,000 rounds) of 7.62mm ammunition has been cycled out of the rotary drum. This shall be repeated until a decision point per Figure 2 is reached.

4.6.4.2 The contractor shall provide replacement parts as required to complete the testing.

4.6.4.3 At the option of the contractor, the test may be interrupted during or at the conclusion of a test cycle, and the test restarted at the point at which it was interrupted. The restart time shall not exceed 65 hours unless approved by the procuring agency.

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. A pilot pack shall consist of a complete rotary drum packaged in accordance with Packaging Data Sheet P8449901 to the level of protection specified in the contract (see 6.2), packed level C and forwarded in accordance with 3.1.

5.2 Preservation, Packaging, Packing and Marking. Rotary drums shall be preserved, packaged, packed and marked in accordance with Packaging Data Sheet P8449901 for the level of protection specified in the contract (see 6.2).

5.3 Repair parts. Repair parts shall be preserved, packaged, packed and marked in accordance with the Packaging Data Sheet or other requirements and for the level of protection specified in the contract (see 6.2).

6. NOTES

6.1 Intended use. The rotary drum is intended for use on, but not limited to, the M28A1 Helicopter Armament Subsystem.

6.2 Ordering data. Procurement documents should specify:

6.2.1 Procurement requirements:

a. Title, number and date of this specification.
b. List of drawings and specifications pertaining to the rotary drum showing applicable revision dates.

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c. Inspection criteria for components (see 4.5.2.1 and 4.5.3.5).
d. Shipping instructions for first article and pilot pack (see 3.1 and 5.1).
e. Reliability and environmental requirements, when required (see 3.5, 3.6 and 4.4).
f. Block of serial numbers (see 3.8).
g. Endurance test lot and sample criteria (see 4.5.1).
h. Disposition of endurance tested drums (see 4.6.2.1).
i. List of inspection equipment, responsibilities for acquisition, maintenance and disposition of, if other than as specified (see 4.5.5.1).
j. Provisions for furnishing the contractor with Government owned parts (see 4.6.1.1).
k. Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).
l. Packaging instructions for repair parts (see 5.3).

6.2.2 Contract data requirements. When warranted, requirements for reports of the results of examination and testing (e.g. functioning, endurance and interchangeability, etc.) shall be specified on a DD Form 1423 included in the contract.

6.2.3 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208 on the Management Control System Summary List, DD Form 1660.

6.3 Unless otherwise specified (see 6.2), the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control System Summary List, DD Form 1660.

6.4 Supersession data. This specification includes the requirements of U.S. Army Weapons Command Purchase Description USAWECOM PD-38, dated 6 November 1972.

Custodian:
Army - WC

Preparing activity:
Army - WC

Project number:
1010-A061

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NOTE: FOR FABRICATION OF THE FIRING TEST STAND ALL STATION, WATERLINE, BUTTLINE DIMENSIONS MARKED ⌀ AND LOCATING DIMENSIONS, OTHER THAN DATUMS [A, B, C, & D] FOR BASIC TURRET AND WEAPON LOCATIONS, APPLY WITH A ± .13 TOLERANCE.

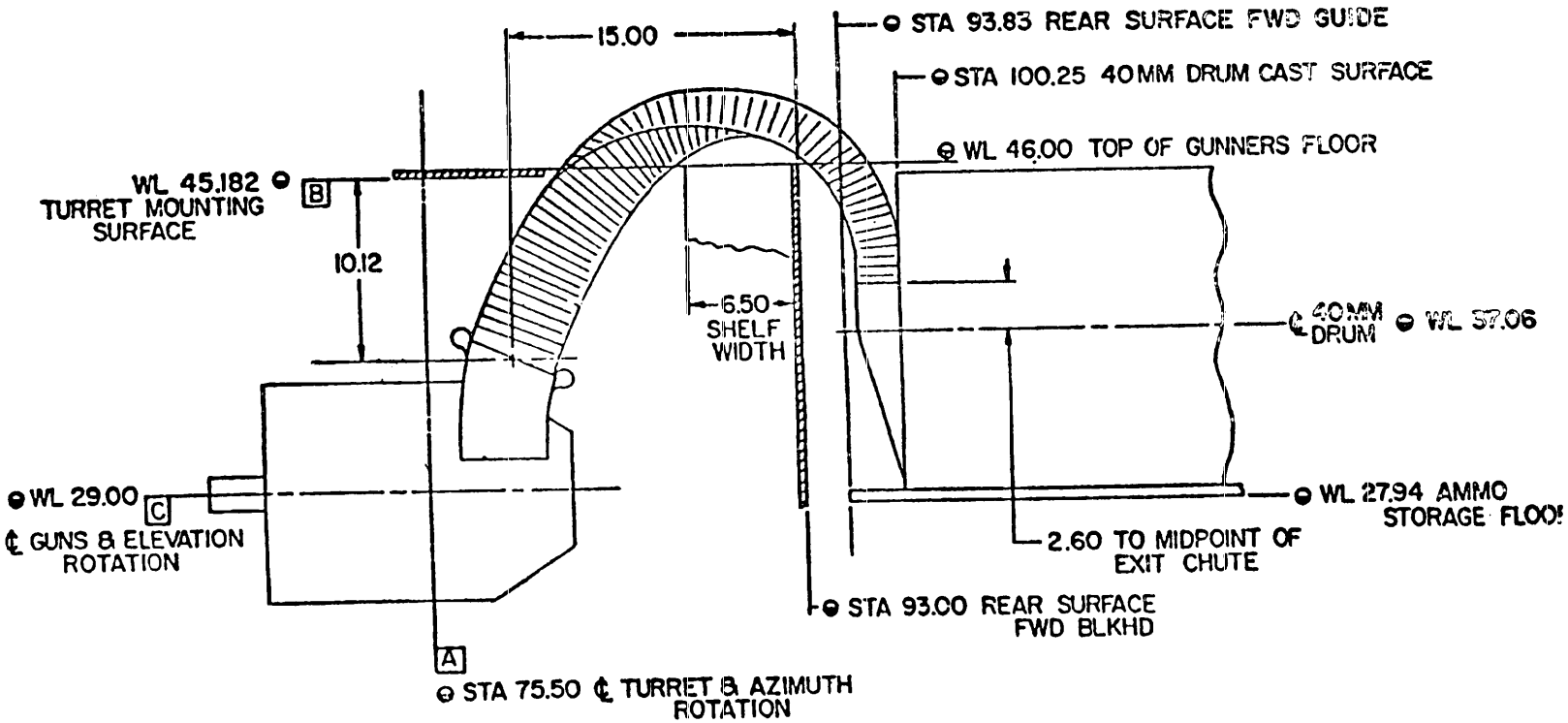
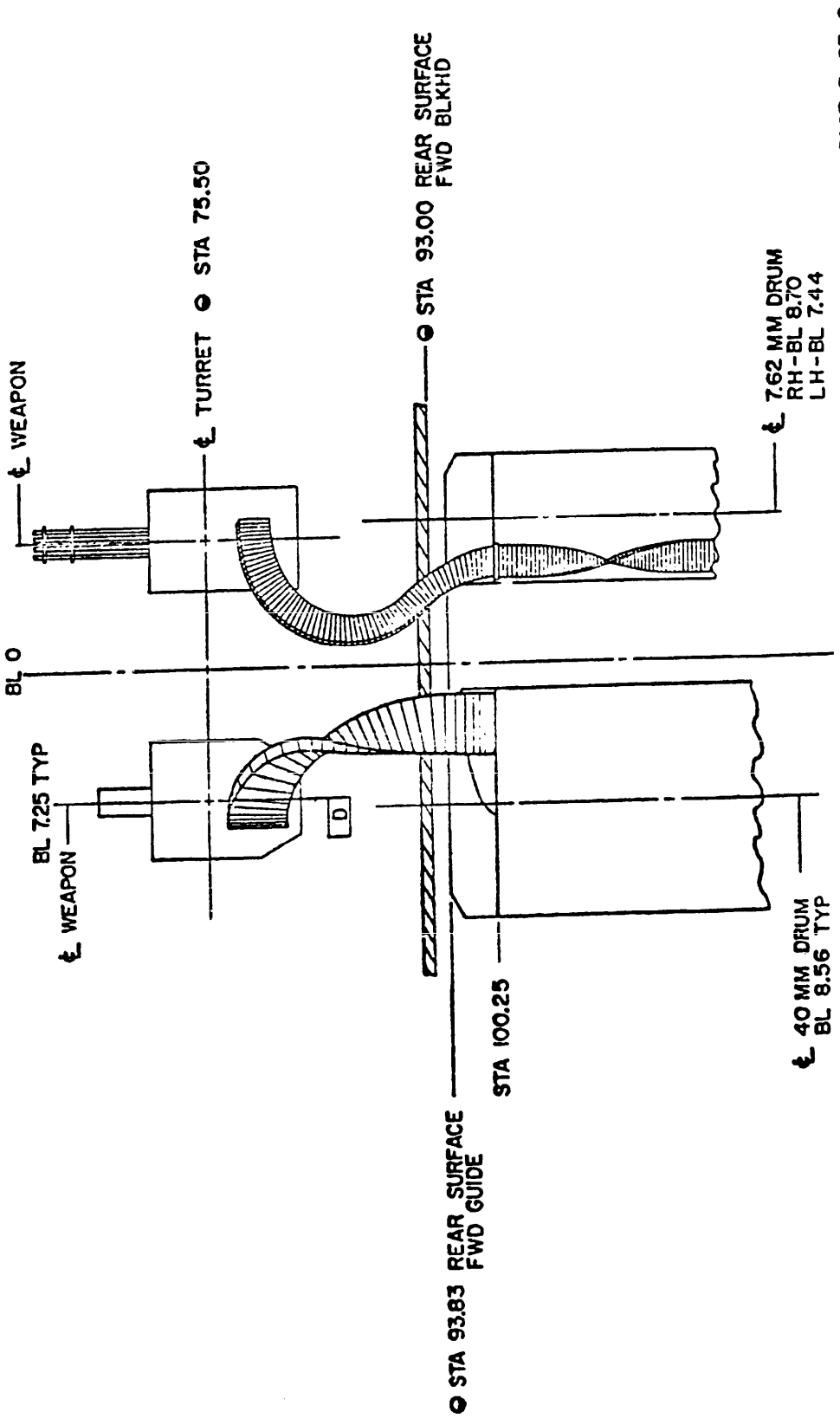


FIGURE 1
RELATIONSHIP OF M28A1 SUBSYSTEM TURRET AND AMMO DRUMS ON AH-1G HELICOPTER.
(40 MM FLEX CHUTE CONF (0° AZ 0° EL POS) SHOWN)

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TOP VIEW SHOWING BUTTLINE LOCATIONS OF WEAPONS & AMMO DRUMS



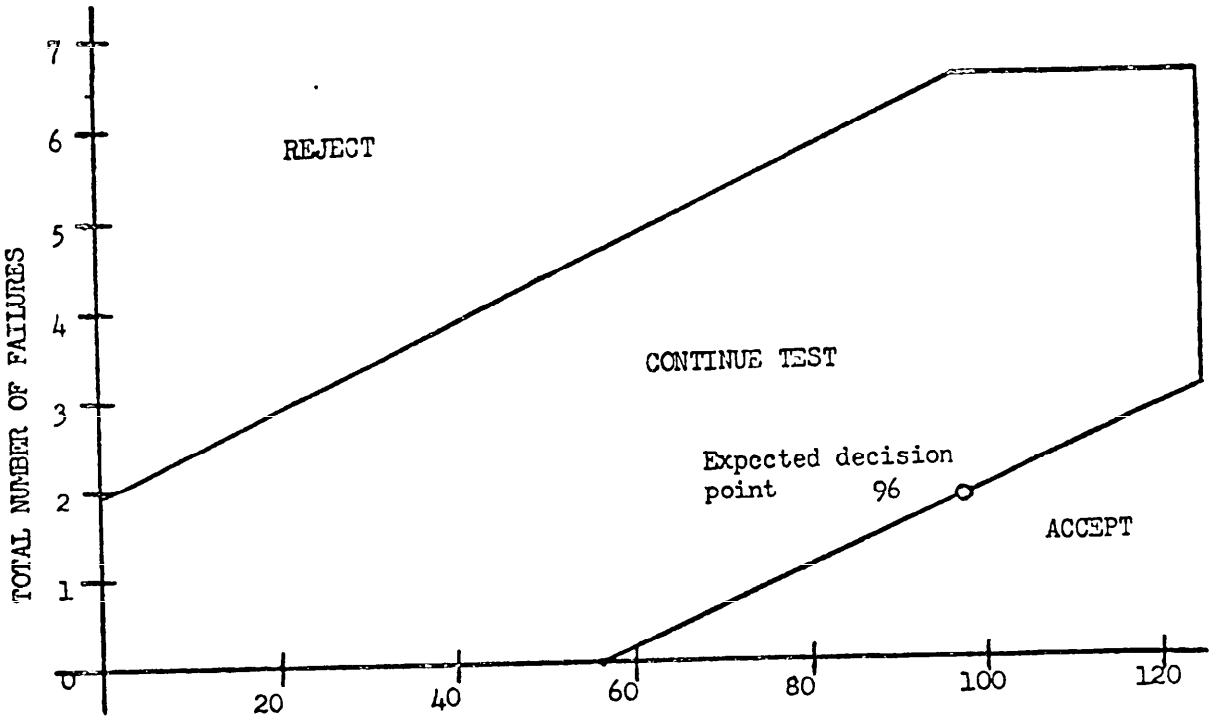
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RELIABILITY ACCEPTANCE TESTING

Decision Risks: 30%

Discrimination Ratio: 1.5:1

MIL-STD-781B - Test Plan VII



TEST CYCLES

No. of failures	Reject (Equal or less)	Accept (Equal or More)
0	N/A	57
1	N/A	79
2	N/A	101
3	22	123
4	44	123
5	66	123
6	123	N/A

FIGURE 2

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-D-45929(WC); Drum, Ammunition, Rotary, 7.62MM

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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